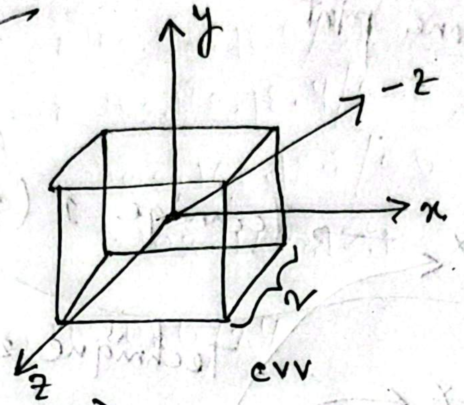
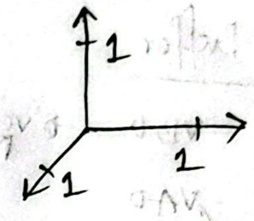
Lab-04

Graphics lab-04

View Fuscum

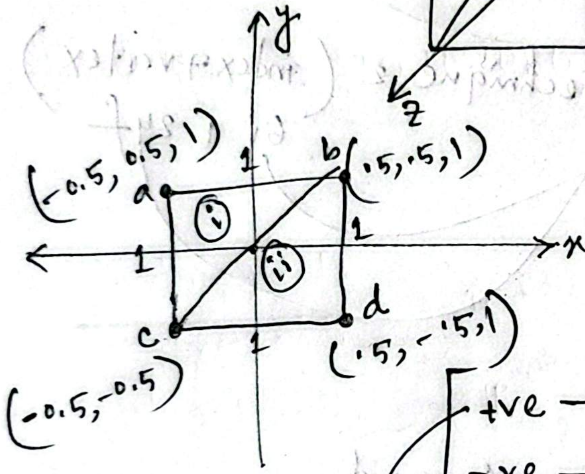




$x \in [-1,1]$

$y \in [-1,1]$

$z \in [-1,1]$


y
(-0.5, 0.5, 1)
a
1
b
(.5, .5, 1)
i
ii
1
1
x
c
1
d
(.5, -.5, 1)
(-0.5, -0.5)

+ve → Anti clockwise grouping ⊙ → frontfacing
-ve → clockwise " ⊗ → backfacing surface

→ কেন দরকার কারণ জানতে হবে

-	1 : a b c
+	2 : c d b

primitive drawing technique

→ everything floating point

Technique 1

1: a b c = 9 points total
$\overset{\uparrow}{xyz}$

2: c d b = 9 " "

18 " "

optimization :

sort unique vertices

index

0 : a
1 : b
2 : c
3 : d

Technique 2

1: 0 1 2 → ~~6~~f + 6i = ~~12~~f
2 : 2 3 1 → ~~0~~f

~~3~~ 4 unique point * 3 dim = 12f

↳ index

12f + 6i

CS CamScanner

$(x, y, z, r, g, b)_a$
└→ 6f for one point

Buffer

VBO → vertex → Techniqe 1 (36f)

VAO

EBO → index

Technique 2 (index + vertex)
6i + 24f

1: abc

2: cdb

→ a b c d e f

A = array $\left[\underbrace{a_x\ a_y\ a_z}_{0}\ \underbrace{a_r\ a_g\ a_b}_{3}\right]$

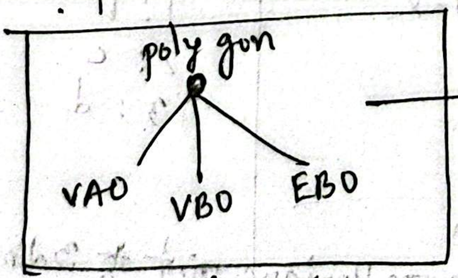
VAO → configuration buffer

	L	offset	stride
cordinate	0	0	6
rgb	1	3	9

stride → কতদূর গেলে আবার back pabo amra

Gourand
└→ x, y, z, r, g, b

Phong
└→ x, y, z, Normal
n_x n_y n_z



→ best করবে

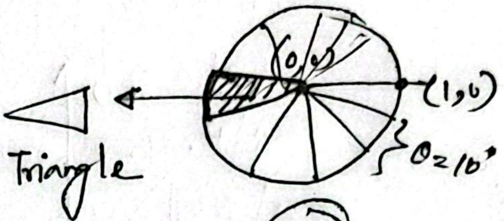
object → চলবে but may drop quality

Hw

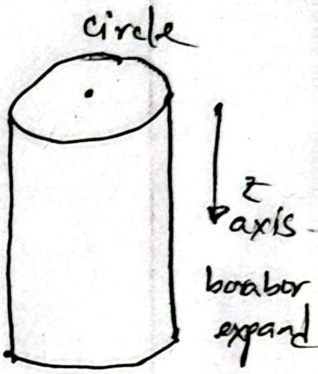
cube, sphere, cylinder

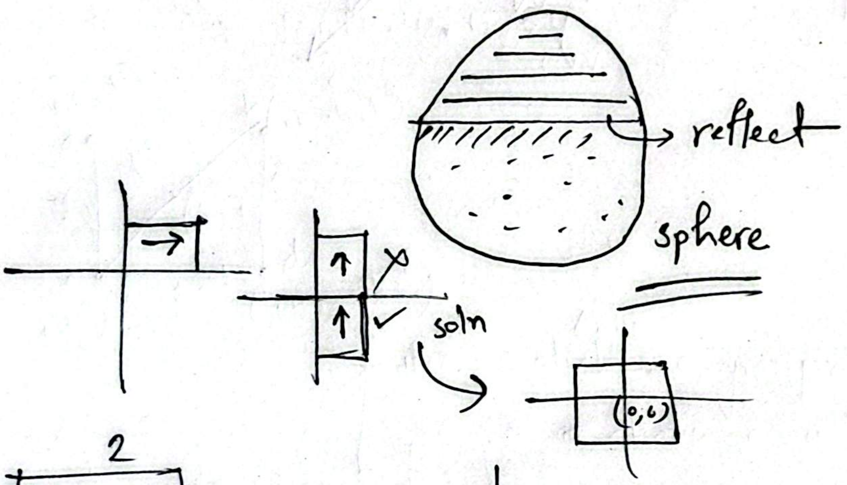
↓

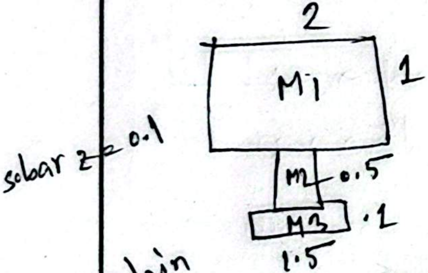
center at origin


Triangle
(0,0)
(1,0)
θ=10°

d = 36   360°/36 = 10°


circle
z axis
borabor expand


reflect
sphere
soln
(0,0)


2
M1
1
sobar z = 0.1
M2
0.5
M3
.1
1.5

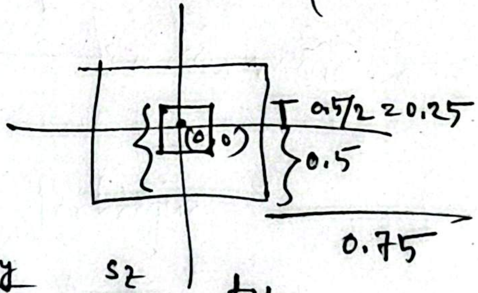

0.5/2 = 0.25
(0,0)
0.5
0.75

Table Maintain kora lagbe →

		Sx	Sy	Sz	ty
M1	=	2	1	.1	
M2	=	0.5	.5	.1	-0.75
M3	=	1.5	1.1	.1	-1.05

-TxRxT

Assignment: $\boxed{\text{All} \rightarrow \text{only object}}$

↳ Excel + explaination lagbe